

Telling Secrets

Secret Writing Through the Ages

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Outline

- 1 Cryptography
- 2 Symmetric Cryptography
 - Transposition Ciphers
 - Substitution Ciphers
- 3 Asymmetric Cryptography
 - Diffie-Hellman Key Exchange
 - RSA Encryption
 - Elliptic Curve Cryptography
- 4 Internet Communication

Cryptography

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Practice of the enciphering and deciphering of messages in secret code in order to render them unintelligible to all but the intended receiver.

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Etymology

- κρυπτός (hidden, secret) from κρύπτω (hide, cover)
- γράφω (write, draw)

... vs. Coding Theory

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Efficient and reliable communication in an uncooperative (possibly hostile) environment.

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Examples

- ASCII, Unicode
- Morse Code
- Universal Product Code (UPC)

Symmetric-Key Algorithms

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Cryptographic algorithms that use the same keys for both encryption and decryption.

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- Alice sends a message to Bob encrypted by a key only she and Bob knows.
- If Eve knows the key, she can decrypt the message.

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Units of plaintext are rearranged to form the ciphertext.

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W . . . S . . . W . . . F . . . T .

- . A . A . H . L . A . S . I . H . S

. . B . . . A . . . Y . . . G . . .

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• Cipher text is “WSWFTA AHLASIH SBAYG”.

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- Plaintext is written on successive rows of fixed length.
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Z E B R A

W A B A S

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Y S F I G

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- Cipher text is “SAGBLFSAASTAWIWHYH”.

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Types

Simple Substitution Cipher: operates on single letters

Polygraphic Cipher: operates on groups of letters

Monoalphabetic Cipher: uses fixed substitution over entire message

Polyalphabetic cipher: uses different substitutions at different positions in the message

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- A becomes D, B becomes E, C becomes F, ..., W becomes Z, X becomes A, Y becomes B, Z becomes C

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- Encrypt "WABASH ALWAYS FIGHTS" with a Caesar right shift of 3.
- A becomes D, B becomes E, C becomes F, ..., W becomes Z, X becomes A, Y becomes B, Z becomes C
- Cipher text is "ZDEDVKDOADBVLJKWV".

Enigma Machine



- Set plugboard and rotors
- Press key on keyboard
- Rotors with randomly wired contacts rotate to new position
- Read letter from the lampboard

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Steps

- Put message in 4×4 column-major order matrix of bytes (the *state*)
- Expand a short key into a number of separate round keys
- AddRoundKey – Combine each byte of the state with the round key using bitwise XOR
- SubBytes – Replace each byte with another from lookup table
- ShiftRows – Shift last three rows cyclically some number of steps
- MixColumns – Combine the four bytes of each column

AES Encryption

- ➊ Expand a short key into a number of separate round keys
- ➋ InitialRound
 - ➊ AddRoundKey – Combine each byte of the state with the round key using bitwise XOR
- ➌ Rounds
 - ➊ SubBytes – Replace each byte with another from lookup table
 - ➋ ShiftRows – Shift last three rows cyclically some number of steps
 - ➌ MixColumns – Combine the four bytes of each column
 - ➍ AddRoundKey
- ➍ Final Round
 - ➊ SubBytes
 - ➋ ShiftRows
 - ➌ AddRoundKey

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- Catch-22

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- Alice and Bob do not need to agree on a key in advance
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- Security based on operations easy to perform, but hard to invert.
- One-way functions described in 1874 by William Jevons.

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 - Relatively easy to compute y given x .
 - Difficult to compute x given y .

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Example

- $p = 101, g = 7$.

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Example

- $p = 101, g = 7$.
- $a = 5$

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Example

- $p = 101, g = 7$.
- $a = 5 \implies A = 7^5 \bmod 101 = 41$

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- $b = 11$

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Example

- $p = 101, g = 7$.
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- $b = 11 \implies B = 7^{11} \bmod 101 = 51$

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- $p = 101, g = 7$.
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- $b = 11 \implies B = 7^{11} \bmod 101 = 51$
- $51^5 \bmod 101 = 60$

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- $p = 101, g = 7$.
- $a = 5 \implies A = 7^5 \bmod 101 = 41$
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- $51^5 \bmod 101 = 60$ and $41^{11} \bmod 101 = 60$.

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- Integer factorization problem.

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- ➏ Alice can encrypt a message M with $\gcd(M, m) = 1$ as $C = M^e \bmod m$.

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 - Solve $ed \equiv 1 \pmod{\varphi(m)}$ by Extended Euclidean Algorithm
- ➏ Alice can encrypt a message M with $\gcd(M, m) = 1$ as $C = M^e \bmod m$.
- ➐ Bob decrypts the ciphertext C as $M = C^d \bmod m$.

RSA Encryption

Example

- $p = 101, q = 103$

RSA Encryption

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- $p = 101, q = 103 \implies m = 10403, \varphi(m) = 10200.$

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- $e = 2243 \implies d = 5507$.

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- Convert "WABASH ALWAYS FIGHTS" to ASCII:

RSA Encryption

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- $e = 2243 \implies d = 5507$.
- Convert "WABASH ALWAYS FIGHTS" to ASCII:
 $M = [87, 65, 66, 65, 83, 72, 32, 65, 76, 87,$
 $65, 89, 83, 32, 70, 73, 71, 72, 84, 83]$

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- $e = 2243 \implies d = 5507$.
- Convert "WABASH ALWAYS FIGHTS" to ASCII:
 $M = [87, 65, 66, 65, 83, 72, 32, 65, 76, 87, 65, 89, 83, 32, 70, 73, 71, 72, 84, 83]$
- Encrypt:
 $C = [2620, 1835, 5420, 1835, 6113, 2874, 544, 1835, 8507, 2620, 1835, 7335, 6113, 544, 1726, 848, 9447, 2874, 1198, 6113]$

RSA Encryption

Example

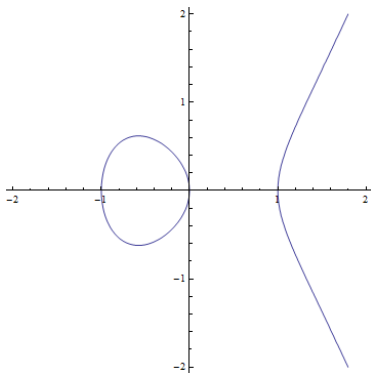
- $p = 101, q = 103 \implies m = 10403, \varphi(m) = 10200$.
- $e = 2243 \implies d = 5507$.
- Convert "WABASH ALWAYS FIGHTS" to ASCII:
 $M = [87, 65, 66, 65, 83, 72, 32, 65, 76, 87, 65, 89, 83, 32, 70, 73, 71, 72, 84, 83]$
- Encrypt:
 $C = [2620, 1835, 5420, 1835, 6113, 2874, 544, 1835, 8507, 2620, 1835, 7335, 6113, 544, 1726, 848, 9447, 2874, 1198, 6113]$
- Decrypt:
 $M = [87, 65, 66, 65, 83, 72, 32, 65, 76, 87, 65, 89, 83, 32, 70, 73, 71, 72, 84, 83]$

Elliptic Curves

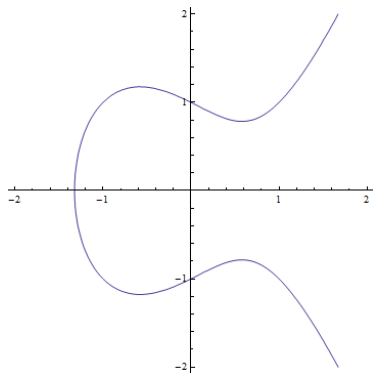
$$y^2 = x^3 + ax + b$$

Elliptic Curves

$$y^2 = x^3 + ax + b$$



$$y^2 = x^3 - x$$



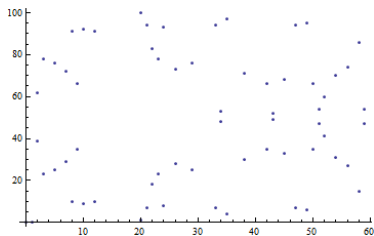
$$y^2 = x^3 - x + 1$$

Elliptic Curves in Finite Fields

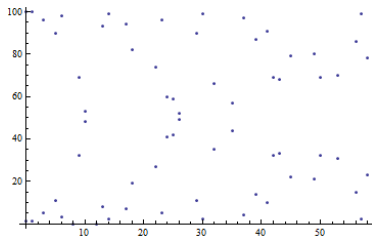
$$y^3 \equiv x^2 + ax + b \pmod{p}$$

Elliptic Curves in Finite Fields

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$$y^2 = x^3 - x$$



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Elliptic Curve Group Structure

- Set of all points on the curve plus infinity

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Elliptic Curve Group Structure

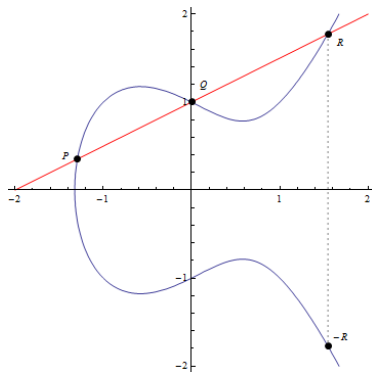
- Set of all points on the curve plus infinity
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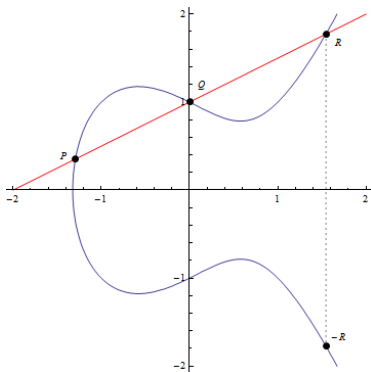


$$P + Q + R = 0$$

$$P + Q = -R$$

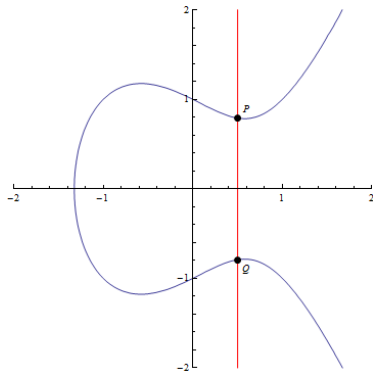
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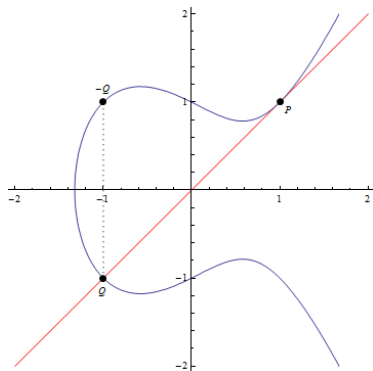
$$P + Q = -R$$



$$P + Q + 0 = 0$$

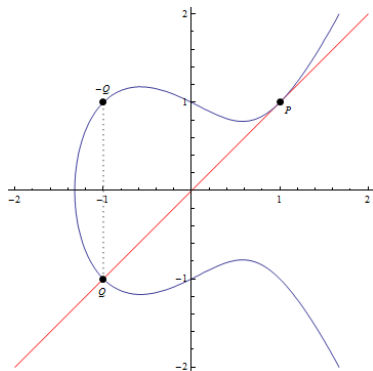
$$P + Q = 0$$

Elliptic Curve Group Operation



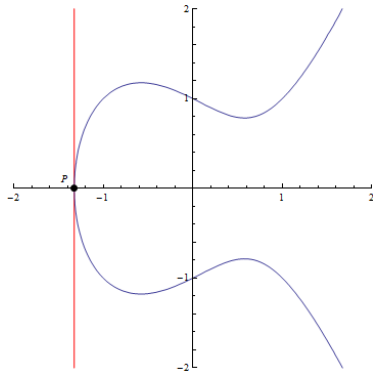
$$P + P + Q = 0$$
$$2P = P + P = -Q$$

Elliptic Curve Group Operation



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$$2P = P + P = -Q$$



$$P + P + 0 = 0$$

$$2P = P + P = 0$$

Elliptic Curve Diffie-Hellman

- Alice and Bob agree on public parameters:
 - Define elliptic curve p, a, b : $y^2 \equiv x^3 + ax + b \pmod{p}$
 - Point on curve G
- Alice picks private integer d_A with $0 \leq d_A < p$.
 - Alice sends Bob $Q_A = d_A G$.
- Similarly, Bob picks d_B and sends Alice $Q_B = d_B G$.
- Alice and Bob compute $(x_k, y_k) = d_A Q_B = d_B Q_A$.
 - Shared secret is x_k .

Transport Layer Security (TLS)

Symmetric Cryptography

- Relatively fast
- Requires shared secret

Asymmetric Cryptography

- Does not require shared secret
- Can authenticate digital signatures
- Typically requires larger keys for same security
- Relatively slow

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- Authenticate certificates (Asymmetric)
 - Establish shared secret (Asymmetric)
 - Conduct session communication in secret (Symmetric)